0.1 Convexity

0.1.1 Definition

Definition 0.1.1. Let $f : (a, b) \to \mathbb{R}$ be a Real-valued function. f is said to be *convex* if: For any $x, y \in (a, b), \lambda \in (0, 1)$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Convex function has follwing properties:

Lemma 0.1.2. Let $f : (a, b) \to \mathbb{R}$ be a Convex function, and $a < x_1 < x_2 < x_3 < b$. Then,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_1)}{x_3 - x_1} \leq \frac{f(x_3) - f(x_2)}{x_3 - x_2}$$

Proof. To show that first inequalty, note that

$$\frac{x_2 - x_1}{x_3 - x_1} \cdot x_3 + \left(1 - \frac{x_2 - x_1}{x_3 - x_1}\right) \cdot x_1 = \frac{(x_2 - x_1)x_3 + (x_3 - x_2)x_1}{x_3 - x_1} = x_2$$

Now,

$$\begin{aligned} f(x_2) &= f\left(\frac{x_2 - x_1}{x_3 - x_1} \cdot x_3 + \left(1 - \frac{x_2 - x_1}{x_3 - x_1}\right) \cdot x_1\right) \\ &\leq \frac{x_2 - x_1}{x_3 - x_1} \cdot f(x_3) + \left(1 - \frac{x_2 - x_1}{x_3 - x_1}\right) \cdot f(x_1) = \frac{x_2 - x_1}{x_3 - x_1} \cdot f(x_3) + f(x_1) - \left(\frac{x_2 - x_1}{x_3 - x_1}\right) \cdot f(x_1) \end{aligned}$$

In brief,

$$f(x_2) - f(x_1) \leq \frac{x_2 - x_1}{x_3 - x_1}(f(x_3) - f(x_1)) \implies \frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_1)}{x_3 - x_1}$$

And similarly,

$$\frac{x_3 - x_2}{x_3 - x_1} \cdot x_1 + \left(1 - \frac{x_3 - x_2}{x_3 - x_1}\right) x_3 = x_2$$

gives the second inequality. □

0.1.2 Properties

Proposition 0.1.3. If $f:(a,b)\to\mathbb{R}$ is Convex, then f is Continuous.

Proof. Let $\varepsilon>0$ be given, $s<t$ are fixed in (a,b). For any $x,y\in(s,t)$ with $s<x<y<t$,

$$\frac{f(s)-f(a)}{s-a}\le\frac{f(x)-f(s)}{x-s}\le\frac{f(y)-f(x)}{y-x}\le\frac{f(t)-f(y)}{t-y}\le\frac{f(b)-f(t)}{b-t}$$

Put $M=\max\left\{\left|\frac{f(s)-f(a)}{s-a}\right|,\left|\frac{f(b)-f(t)}{b-t}\right|\right\}$. Then, for any $x,y\in(s,t)$,

$$\left|\frac{f(y)-f(x)}{y-x}\right|\le M$$

Now,

$$|y-x|<\frac{\varepsilon}{M}\implies |f(y)-f(x)|\le M|y-x|<\varepsilon$$

Since $s,t\in(a,b)$ was arbitrary, f is continuous on (a,b). □

Proposition 0.1.4. Let f is differentiable on (a,b). Then,

$$f \text{ is Convex } \textit{if and only if } f' \text{ is monotonically increasing on } (a,b).$$

Proof. Prove by showing both directions: right and left.
Right Direction Let $x_1<x_2$ in (a,b). Then,

$$f'(x_1)=\lim_{t\to x_1}\frac{f(t)-f(x_1)}{t-x_1}\le\frac{f(x_2)-f(x_1)}{x_2-x_1}\le\lim_{\tau\to x_2}\frac{f(\tau)-f(x_2)}{\tau-x_2}=f'(x_2)$$

More rigorously, put $\varepsilon=\left|f'(x_1)-\frac{f(x_2)-f(x_1)}{x_2-x_1}\right|$. (If $\varepsilon=0$, then there is nothing to prove.).
Now, there exists a $\delta>0$ such that $|t-x_1|<\delta$ implies

$$\left|f'(x_1)-\frac{f(t)-f(x_1)}{t-x_1}\right|<\varepsilon\iff-\varepsilon+\frac{f(t)-f(x_1)}{t-x_1}<f'(x_1)\overset{(*)}{=}\varepsilon+\frac{f(t)-f(x_1)}{t-x_1}$$

If $f'(x_1)-\frac{f(x_2)-f(x_1)}{x_2-x_1}>0$, then $(*)$ gives

$$f'(x_1)<f'(x_1)+\frac{f(t)-f(x_1)}{t-x_1}-\frac{f(x_2)-f(x_1)}{x_2-x_1}\iff\frac{f(x_2)-f(x_1)}{x_2-x_1}<\frac{f(t)-f(x_1)}{t-x_1}\quad\forall t\ \text{ s.t }\ |t-x_1|<\delta$$

If $|t-x_1|<|x_2-x_1|$, then this contradicts to Convexity.
Consequently, we obtain the first inequality, similarly can prove the second inequality.
Left Direction Let $x,y\in(a,b)$ and $\lambda\in(0,1)$ be given. The Mean Value Theorem gives that:

$$f(\lambda x+(1-\lambda)y)-f(x)=f'(z_1)(\lambda x+(1-\lambda)y-x)\ \text{ for some }\ z_1\in(x,\lambda x+(1-\lambda)y)$$
$$f(y)-f(\lambda x+(1-\lambda)y)=f'(z_2)(y-\lambda x+(1-\lambda)y)\ \text{ for some }\ z_2\in(\lambda x+(1-\lambda)y,y)$$

Now, Monotonically increasing gives

$$\begin{aligned}&\frac{f(\lambda x+(1-\lambda)y)-f(x)}{\lambda x+(1-\lambda)y-x}=f'(z_1)\le f'(z_2)=\frac{f(y)-f(\lambda x+(1-\lambda)y)}{y-(\lambda x+(1-\lambda)y)}\\ \implies&\frac{f(\lambda x+(1-\lambda)y)-f(x)}{(1-x)(y-x)}\le\frac{f(y)-f(\lambda x+(1-\lambda)y)}{\lambda(y-x)}\\ \implies&\lambda f(\lambda x+(1-\lambda)y)-\lambda f(x)\le(1-\lambda)f(y)-(1-\lambda)\lambda x+(1-\lambda)y\\ \implies&f(\lambda x+(1-\lambda)y)\le\lambda f(x)+(1-\lambda)f(y)\end{aligned}$$

□

Corollary 0.1.5. If $f:[a,b]\to\mathbb{R}$ is twice-differentiable, then

$$f \text{ is Convex } \textit{if and only if } f''(x)\geq 0 \text{ for all } x\in(a,b).$$

Theorem 0.1.6. Let $f:[a.b]\to\mathbb{R}$ be given. Then,

$$f \text{ is Convex if and only if } f \text{ is Continuous, and Midpoint Convex.}$$

Midpoint convex is that f satisfies $\forall x,y\in(a,b),\ f\left(\frac{x+y}{2}\right)\leq\frac{f(x)+f(y)}{2}$.

Proof. The right direction is clear. To show the left direction, we demonstrate that *Midpoint Convexity implies Dyadic Rational Convexity*. Claim: For any $n\in\mathbb{N}$,

$$f\left(\frac{\sum_{k=1}^{2^n}x_k}{2^n}\right)\leq\frac{1}{2^n}\sum_{k=1}^{2^n}f(x_k) \tag{$*$}$$

Using Induction: If $n=1$, it is clear by Midpoint Convexity.
Assume that for $n\in\mathbb{N}$, $(*)$ is True. Then,

$$\begin{aligned}
f\left(\frac{\sum_{k=1}^{2^{n+1}}x_k}{2^{n+1}}\right) &= f\left(\frac{1}{2}\cdot\left[\frac{\sum_{k=1}^{2^n}x_k}{2^n}+\frac{\sum_{k=2^n+1}^{2^{n+1}}x_k}{2^n}\right]\right)\\
&\overset{\text{m.c}}{\leq}\frac{1}{2}\left(f\left(\frac{\sum_{k=1}^{2^n}x_k}{2^n}\right)+f\left(\frac{\sum_{k=2^n+1}^{2^{n+1}}x_k}{2^n}\right)\right)\\
&\overset{(*)}{\leq}\frac{1}{2}\left(\frac{1}{2^n}\sum_{k=1}^{2^n}f(x_k)+\frac{1}{2^n}\sum_{k=2^n+1}^{2^{n+1}}f(x_k)\right)\\
&=\frac{1}{2^{n+1}}\sum_{k=1}^{2^{n+1}}f(x_k)
\end{aligned}$$

Consequently, we obtain the claim. Now, let $n\in\mathbb{N}$, and m be an integer such that $1\leq m\leq 2^n$.
Put $x_1=x_2=\cdots=x_m=x$ and $x_{m+1}=x_{m+2}=\cdots=x_{2^n}=y$. Then

$$f\left(\frac{m}{2^n}x+\left(1-\frac{m}{2^n}\right)y\right)\leq\frac{m}{2^n}f(x)+\left(1-\frac{m}{2^n}\right)f(y)$$

For complete this discussion, Let $x,y\in(a,b),\lambda\in(0,1)$ be given.
Since $\frac{\lfloor 2^n\lambda\rfloor}{2^n}\to\lambda$ as $n\to\infty$, for any $n\in\mathbb{N}$,

$$f\left(\frac{\lfloor 2^n\lambda\rfloor}{2^n}x+\left(1-\frac{\lfloor 2^n\lambda\rfloor}{2^n}\right)y\right)\leq\frac{\lfloor 2^n\lambda\rfloor}{2^n}f(x)+\left(1-\frac{\lfloor 2^n\lambda\rfloor}{2^n}\right)f(y)$$

Finally, taking limits then

$$\lim_{n\to\infty}f\left(\frac{\lfloor 2^n\lambda\rfloor}{2^n}x+\left(1-\frac{\lfloor 2^n\lambda\rfloor}{2^n}\right)y\right)\overset{f\ \text{conti.}}{=}f\left(\lim_{n\to\infty}\left[\frac{\lfloor 2^n\lambda\rfloor}{2^n}x+\left(1-\frac{\lfloor 2^n\lambda\rfloor}{2^n}\right)y\right]\right)=f(\lambda x+(1-\lambda)y)\leq\lambda f(x)+(1-\lambda)f(y)$$

In sum, Midpoint Convexity implies Dyadic Rational Convexity, and with Continuous implies Convexity. □

Let $f: (0,1) \to \mathbb{R}$ be a convex map.

1). For any $0<x<y<1$, $f'_+(x) \le f'_-(y)$.

2). The set $\{x \in (0,1) \mid f \text{ is not differentiable at } x\}$ is at most Countable.

Proof. First, Since f is convex, for any $x_1<x_2<x_3$ in $(0,1)$,

$$\frac{f(x_2)-f(x_1)}{x_2-x_1} \le \frac{f(x_3)-f(x_1)}{x_3-x_1} \le \frac{f(x_3)-f(x_2)}{x_3-x_2}$$

1). $f_+(x)$ exists for any $x \in (0,1)$

Let $x \in (0,1)$ be given. And, let $a<x$ be fixed.

Then, for any $0<h$,

$$\frac{f(x)-f(a)}{x-a} \le \frac{f(x+h)-f(x)}{h},$$ and since $\frac{f(x+h)-f(x)}{h}$ decreases as $h \to 0$,

so $f'_+(x)$ exists. Similarly $f'_-(x)$ exists.

2). $x<y \Rightarrow f_+(x) \le f_-(y)$.

Take $\delta = \frac{1}{2}|x-y|$. Then for all $h \in \mathbb{R}$ with $0<h<\delta$,

$$\frac{f(x+h)-f(x)}{x+h-x} < \frac{f(y)-f(y-h)}{y-(y-h)},$$

So the first claim proved.

Now, let $x \in X$ be a point such that f is not differentiable.

Since $f'_+(x)$ and $f'_-(x)$ exist, the only possibility is

$$f'_-(x) < f'_+(x).$$

By the density of Rationals there is $r \in (f'_-(x), f'_+(x)) \cap \mathbb{Q}$.

And, for any $x<y$, $f'_+(x) \le f'_-(y)$, if y is another not differentiable point, $x<y$, then $r_x < r_y$, so the map from the set into $\mathbb{Q}$: $x \mapsto r_x$ is injective.

So, the set is at most Countable.